ParkFlow

A FASTag and ANPR Fusion Platform for Frictionless, "Just Drive" Parking Across India

Section 1. Title of the Project / Product

Project Name: ParkFlow

Full Descriptor: A FASTag and ANPR Fusion Platform for Frictionless, "Just Drive" Parking Across India

Positioning Line: The walk out store experience, applied to parking. Vehicles are recognised, verified and billed automatically as they pass, with no stop, no token and no manual payment.

Sector: Smart Mobility / Urban Infrastructure Technology

Core Stack: RFID (FASTag / NETC), ANPR (computer vision), on site edge compute, IoT occupancy sensing, cloud billing and analytics **Target Segments:** Tech parks and corporate campuses, malls and retail, residential complexes (RWAs), government and municipal parking, and any other private managed lot

Section 2. Problem Statement

Parking across India's malls, tech parks, residential complexes and government facilities still runs on manual tokens, boom barriers and cash counters, forcing every vehicle to stop and queue at both entry and exit, even though close to 98 percent of those same vehicles already carry a government issued, uniquely identifiable RFID tag that goes completely unused the moment they leave the highway.

This single gap compounds into three distinct, measurable problems for every stakeholder in the parking value chain:

- **Congestion and lost time.** Manual token issuance and cash or card settlement typically hold a vehicle at the barrier for 45 to 90 seconds. At peak hours, such as mall weekends and tech park shift changes, this compounds into queues that spill onto public roads.
- **Revenue leakage and fraud.** Paper tokens can be duplicated, under reported or sold off book by operating staff, a well documented pain point in cash heavy Indian parking operations, and one that is especially acute for municipal and government run lots with limited audit capacity. **Poor visibility and weak safety data.** Most facilities have no real time record of how many vehicles are on site, where they are, or who they belong to, a gap that matters for everyday security as well as for emergency evacuation planning.

The insight: India does not need a new piece of plastic, a new sticker or a new wallet to fix this. It needs the identity and settlement rail it already built for highways, FASTag on NETC, extended, safely and redundantly, into the country's parking gates.

Section 3. Solution Proposed

ParkFlow is a site side compute and cloud software platform that reads two independent signals from every approaching vehicle: its FASTag RFID identity and its number plate. It cross verifies them on an on premise compute node in real time, and uses that fused decision to open the lane, or keep it open, guide the driver to a free slot, and settle the fee automatically through the driver's existing FASTag linked wallet. It follows the same operating principle as a modern cashier less store: sensors identify what is moving through the space, a local decision engine resolves identity with high confidence, and billing happens invisibly in the background.

3.1 How it works

1. The vehicle approaches at normal driving speed. No braking is required.
2. Lane sensors, a UHF RFID reader and an ANPR camera, read the tag and the plate at the same time.
3. The ParkFlow Edge Node, described in Section 3.3, fuses both signals and reaches a decision in under 300 milliseconds.
4. On a confident match, four things happen at once: the barrier opens or the lane stays free; the driver's app and wallet

receive an automatic debit and an e receipt; the operator dashboard logs live occupancy, revenue and the audit trail; and IoT guidance directs the driver, by LED signage and the app, to the nearest open slot.

Low confidence or no match cases, such as an unregistered visitor, a damaged tag or an unreadable plate, fall back to a 2 to 3 second guard assisted visual check or an in app QR check in. The system never hard blocks a vehicle without a human reviewable alternative.

3.2 Core features

Feature	What it does	Why it matters
Dual sensor fusion	Requires a FASTag RFID read and an ANPR plate read to agree before granting a barrier free pass.	Closes the two biggest single sensor failure modes: tag cloning or misuse, and plate misreads from non standard, dirty or two wheeler plates.
Barrier optional lanes	The boom barrier auto lifts before arrival for verified vehicles, or is removed entirely for closed campuses.	Converts a stop and go bottleneck into continuous flow.
NETC wallet settlement	Debits the fee from the same prepaid FASTag wallet already used for tolls.	Zero onboarding friction. No new app top up or QR scan for a returning driver.
Companion app	Live slot availability, pre booking, visitor QR passes, monthly resident passes, EV bay reservation.	Extends the experience to visitors and non FASTag vehicles without breaking the flow.
IoT occupancy and guidance	Per zone sensors or camera analytics feed LED signage and the app with real time free slot direction.	Eliminates the time spent circling for a spot, which dominates dense basements and multi level structures.
Dynamic pricing engine	Rule based and machine learning based pricing: peak surcharge, resident subscriptions, validated or free shopper parking.	Lets each operator monetise, or subsidise, parking on their own terms.
Operator analytics dashboard	Occupancy heatmaps, revenue reconciliation, staffing forecasts, and a full digital audit trail.	Replaces manual cash counting and closes leakage.

3.3 Site side compute: the ParkFlow Edge Node

Every site gets one on premise compute unit, the ParkFlow Edge Node, mounted in the facility's existing server or security room. All plate recognition and identity fusion happen on this unit, not in the cloud. This is a deliberate architectural choice, and it carries most of the system's reliability, latency and privacy properties.

Capability	Purpose
Local ANPR inference	Runs the plate recognition neural network on an on board accelerator, returning a plate read in roughly 60 to 120 milliseconds. No video needs to leave the site for a routine decision.
Identity fusion and access decision	Correlates the RFID tag read with the plate read, checks the local access list of

	employees, residents, pre booked visitors and the after hours pool, and drives the barrier. The full loop completes on site in under 300 milliseconds.
Offline first operation	The site keeps working through an internet outage. Entries and exits are queued locally and settled against NETC once the link returns. Connectivity becomes a billing dependency, not an access dependency, which matters for basement lots and tier 2 sites.
Privacy by data minimisation	Raw video and plate crops are processed and discarded locally. Only a tokenised vehicle reference and the transaction record sync to the cloud, keeping the system aligned with data minimisation expectations under India's data protection framework rather than streaming footage off site by default.
Multi lane and multi sensor host	A single node serves roughly 4 to 8 lanes plus the site's occupancy sensors and LED signage, so hardware cost amortises across the whole facility instead of scaling per lane.
Over the air model updates	Recognition models improve over time as the fleet encounters more Indian plate variants, including non standard fonts, decorative plates, state formats and two wheeler layouts, with no site visit or hardware swap required.
Local audit buffer	Retains a rolling on site log of entries and exits, so the operator's audit trail survives a cloud or network incident.

Why edge, not cloud. A cloud round trip for every gate event would add network latency to a decision that has to finish before a moving vehicle reaches the barrier, would fail closed whenever the site's link drops, and would require streaming number plate imagery off premise. Putting the compute on site removes all three problems at once, and, as Section 6 sets out, the same node later doubles as the merchant terminal for non parking transactions.

3.4 Technology stack

- **Hardware:** UHF RFID lane reader, ANPR camera, the ParkFlow Edge Node (an accelerated on site compute unit), IoT occupancy sensors (ultrasonic or camera based), LED guidance signage, and an optional boom barrier that is retrofit compatible with most existing barriers.
- **Software:** on node ANPR inference and fusion or matching engine, cloud parking management platform, driver mobile app for iOS and Android, and an operator web dashboard.
- **Integrations:** the NETC / NPCI merchant acquiring API for FASTag wallet debits, UPI as a fallback payment rail, smart city and municipal ERP APIs, and third party navigation and ride hailing APIs for live availability.

Section 4. Implementation Plan

4.1 Target beneficiaries

Drivers and residents: daily commuters into tech parks and malls, residents and families in gated complexes, and visitors, delivery riders and cab or fleet drivers.

Facility owners and operators: mall management companies, tech park and SEZ developers and corporates, and Resident Welfare Associations.

Government and city bodies: municipal corporations and Smart City special purpose vehicles, traffic police using congestion and safety data, and airport, metro and railway parking authorities.

4.2 Phased rollout

Phase	Timeline	Scope	Milestone

Phase 0, Pilot	Months 1 to 3	Two tech parks and one mall in one metro, for example Bengaluru, Hyderabad or Pune, chosen for tech park density and digital first users.	Hardware live on three sites. NETC integration validated end to end.
Phase 1, Prove Interoperability	Months 4 to 9	25 to 30 sites across three to four metros. Add residential (RWA) complexes and one government run lot through a municipal memorandum of understanding.	Public, private and residential ownership models all live on one platform.
Phase 2, Pan India Scale	Months 10 to 18	150 or more sites across the top 8 to 10 cities. EV bay booking, fleet and corporate billing, and navigation app API partnerships.	Multi city network effect. The platform becomes the default rail for new client onboarding.

4.3 Deliverables per site

Lane hardware kit: RFID reader, ANPR camera, and boom barrier integration or a barrier less signage kit.

The ParkFlow Edge Node, installed and commissioned, serving all lanes at the facility.

Cloud onboarding and NETC merchant configuration.

Operator dashboard access and staff training.

Driver facing app listing for the site, with resident, employee and visitor tiers configured.

Service level agreement backed uptime monitoring and support.

4.4 Expected outcomes (pilot validated targets)

Metric	**Today (typical baseline)**	**ParkFlow target**
Time per vehicle at entry or exit	45 to 90 seconds	Under 5 seconds for verified vehicles
Peak hour queue length	10 to 20 or more vehicles at busy sites	Near zero. The barrier opens ahead of arrival.
Revenue leakage (cash or token lots)	Meaningful, operator reported leakage	Near zero. Every transaction is digitally logged.
Digital audit trail coverage	Little to none	100 percent of entries and exits time stamped and logged

Figures above are internal projections to be validated during the Phase 0 pilot, not third party audited results.

4.5 Key risks and mitigations

Risk	**Mitigation**
Sensor misread, such as dirty or non standard plates, or tag damage	Dual sensor fusion plus a graceful fallback to QR or app check in or a guard review. The system never issues a hard block.
Site connectivity gaps	An edge first architecture. Matching happens on device and syncs to the cloud once connectivity resumes.

Resistance from token vendors or security staff	A phased co existence period, and reskilling on site staff into verification and customer assistance roles.
Data privacy and regulatory compliance	Encrypted storage, consent based data sharing, alignment with the IT Act and the Digital Personal Data Protection Act, and law enforcement data shared only on lawful request.

Section 5. Safety and Efficiency Analysis

5.1 Safety

Fraud and tailgating resistance. A vehicle without a valid, KYC verified FASTag and a matching plate is flagged rather than waved through, unlike a lost or duplicated paper token, which grants access to anyone holding it.

Full evidentiary trail. Every entry and exit is time stamped, photographed and logged, supporting stolen vehicle recovery and incident investigation, shared with law enforcement only through lawful, consent based channels.

Less human exposure at the barrier. Guards no longer handle cash or tokens in live traffic lanes, removing a routine source of minor barrier side accidents. **Emergency readiness.** A live, per level occupancy count gives facility managers an accurate real time headcount of vehicles on site, directly useful for fire and emergency evacuation planning, something manual token systems cannot provide.

Less circling, fewer in lot conflicts. Guided routing to a known free slot reduces reversing, idling and searching, the main causes of low speed in lot collisions.

5.2 Efficiency

Throughput math. Cutting the service time at a barrier from roughly 60 seconds to roughly 2 to 3 seconds multiplies the effective vehicles per hour capacity of a single lane by roughly twenty times. This is the single largest lever in the whole system.

Less circling, less fuel burnt. Real time guidance to a specific free slot removes the multi minute search common in large basements and multi level structures.

Leak proof reconciliation. Automated, digital billing removes manual cash counting, a widely cited source of leakage in cash based Indian parking operations, and gives operators same day, auditable revenue reports.

Predictive staffing. Historical and live occupancy data lets operators right size security and valet staffing around known peaks, such as mall sales, tech park shift changes and festival footfall.

Dimension	**Manual token system**	**ParkFlow**
Vehicle identity check	Physical token only. Transferable and forgeable.	RFID and ANPR fusion. Non transferable and dual verified.
Cash handling	Manual, staff exposed	None. Automatic wallet debit.
Occupancy visibility	Manual counts, often stale	Real time, per zone and per level
Audit trail	Paper based or absent	100 percent digital and time stamped
Effective lane throughput	Roughly 40 to 60 vehicles per hour	800 to 1,000 or more vehicles per hour, indicative, to be confirmed in the pilot

Section 6. Scalability and Future Development

ParkFlow is designed to grow along three horizons: deepening in its core segments first, then broadening into adjacent verticals and geographies, and finally opening into a platform that other products build on.

Horizon	Timeframe	Focus	Key moves
Horizon 1, Core	0 to 12 months	Win the three founding segments in top metros	Tech parks, malls and residential complexes across 8 to 10 metros. Harden the fusion accuracy and the NETC billing pipeline.
Horizon 2, Adjacent	12 to 30 months	New verticals and geographies	Airports, hospitals, stadiums and on street municipal, or metered, parking using a camera only pole mounted variant. Tier 2 city expansion through a capex light Build Operate Transfer, or franchise, model. EV charging bay discovery and booking. Linkage to the National Common Mobility Card for combined metro and parking fare journeys.
Horizon 3, Platform	30 or more months	Open ecosystem and data products	An open availability and booking API for Maps and ride hailing apps. Anonymised, aggregated mobility and footfall analytics as a data product for urban planners and retail. A single, interoperable network where one tag works at any parking lot nationwide. Adjacent extensions into highway rest stop parking and logistics or fleet yard management.

Why this scales cheaply. ParkFlow rides on infrastructure the country has already paid for and mandated: FASTag issuance, KYC and the NETC settlement network. Each new site adds lane hardware and a software onboarding step, not a new identity system, a new bank tie up, or a new government mandate. That keeps marginal rollout cost, and time to revenue, low even in tier 2 and tier 3 cities.

6.1 Commercialising private parking: unlocking idle capacity

A typical tech park's parking structure is a large, expensive asset that earns nothing for roughly two thirds of the week. It fills between about 9 AM and 7 PM on weekdays and sits close to empty every evening, weekend and public holiday, often while restaurants, cinemas, event venues and under parked residential streets a few hundred metres away are turning cars away. The reverse pattern holds inside gated residential complexes, where visitor bays sit unused all day while residents are at work.

The reason this arbitrage has never been captured at scale is not demand, it is access control. No facility manager will open a private, security sensitive campus to the general public when the only gate technology available is a paper token that any driver can hold. Verified vehicle identity is what makes it safe, and that is exactly what the Edge Node already provides at every ParkFlow site.

Asset	Idle window	Monetisation ParkFlow enables
Tech parks and corporate campuses	Weekday evenings, weekends, holidays	Release a ring fenced block of bays to a public "after hours pool" on time boxed, geofenced access rules, priced dynamically against nearby demand, with tenant bays untouched and every entrant identity verified.
Malls and retail	Overnight, weekday mornings	Monthly overnight subscriptions for residents of surrounding buildings that lack off street parking, and early morning capacity for gyms and clinics.
Residential complexes	Weekday daytime (visitor bays)	Daytime hourly parking sold to nearby offices and shoppers, with an RWA set cap on how many outside vehicles may be inside at once.

Government and municipal lots	Outside office hours	Evening and weekend public parking, converting a fixed cost civic asset into a revenue line without new construction.

The commercial mechanism. Idle bays are listed on the ParkFlow network with owner defined rules: which hours, how many bays, which zones, and the minimum price. Drivers discover and pre book them in the app. Access is granted by the same tag plus plate check used for tenants, so the owner never issues a credential to a stranger. Revenue is split between the facility owner and ParkFlow, which turns the platform from a cost centre the operator pays for into a yield line the operator earns from, a materially easier sale, and one that shortens payback on the hardware. The owner retains a hard override: a single dashboard control closes the pool instantly.

6.2 Beyond parking: the vehicle as a payment credential

Once a site has an Edge Node that can identify a vehicle with high confidence and debit its NETC wallet, that node is no longer only a parking controller, it is a merchant terminal for any transaction where a vehicle is the customer. The installed base built for parking becomes the acquiring network for vehicle commerce.

Use case	**How it works and what it removes**
Fuel stations	The vehicle is identified as it pulls onto the forecourt. The attendant fills, the amount is confirmed on the pump side terminal, and the wallet is debited. This removes cash handling and card machine queuing at one of India's highest frequency retail interactions, and the plate plus tag check guards against tag misuse on a higher value transaction than parking.
Drive throughs (quick service restaurants)	The vehicle is recognised on approach to the order point, so the order is tied to it and payment settles while the car moves between the speaker and the window. This cuts the window handover to collection only, which is the single largest constraint on drive through throughput.
EV charging	Plug and charge style billing. The bay identifies the vehicle, starts the session and settles automatically, with no separate app or RFID card per charging network.
Car washes, service centres, valet	Booking, check in and payment collapse into a single drive in event, with the service history tied to the verified vehicle rather than a phone number.
Logistics yards and weighbridges	Gate in and gate out automation for freight, with the same identity used for demurrage billing and yard dwell time analytics.

Honest dependency: this is a Horizon 2 or 3 ambition, not a launch feature. FASTag based fuel payment already exists in the Indian market and is offered by at least one incumbent, so ParkFlow does not claim novelty on the use case itself. What is defensible is the route: a fleet of already deployed, identity verifying Edge Nodes and one merchant network across categories, rather than a category by category integration. Realising it depends on NPCI merchant category approvals and acquirer or oil marketing company partnerships, which are outside ParkFlow's unilateral control. The plan therefore treats parking as the standalone business case and vehicle commerce as optionality layered on the same hardware. The proposal does not rely on these approvals to be viable.

Section 7. Innovation Advantage

India's parking technology space is not empty, and ParkFlow's positioning is built with that fully in view.

7.1 Competitive landscape

Player type	Focus	Key Point
Legacy token or boom barrier vendors	Hardware only, regionally fragmented, no digital identity or billing layer	Still the majority install base across Indian malls and tech parks today
Park+	A car ownership "super app" offering FASTag, ANPR and parking services alongside challans, insurance and car care	Founded in 2019. Has raised over 50 million US dollars from investors including Sequoia India and Matrix Partners.
Get My Parking / ParkingRhino	IoT based smart parking platforms for municipalities, airports, universities and corporates	Delivered smart city parking pilots with municipal bodies including in Jaipur and Navi Mumbai
City specific aggregators, for example GetParking	ANPR based off street parking aggregation in select cities	Operate a network of parking lots aggregated primarily in a handful of metros

Based on public company information as of the most recent public disclosures. Figures are self reported by the respective companies and are indicative only.

7.2 Where ParkFlow differentiates

Fusion, not either or. Most deployments in the market today lean on a single modality, RFID or ANPR, per site. ParkFlow's edge engine requires both to agree before granting a stop free pass, and degrades gracefully, through visual confirmation or a QR fallback, rather than hard failing, pushing accuracy and driver experience above single sensor deployments.

Zero friction settlement. Billing through the driver's existing NETC linked FASTag wallet, the same one they already top up for tolls, removes the first visit friction of installing a new app or topping up a new proprietary wallet.

One identity, every segment. A single vehicle identity works across tech park, mall, residential and government lots in the same city, enabling a "park anywhere" network effect and unified billing for corporates managing multi site employee parking.

A capex light path into the underserved public sector. A revenue share or Build Operate Transfer model designed specifically for cash strapped municipal and RWA managed lots, a segment long municipal sales cycles keep many private players from prioritising, and exactly where ParkFlow's phased, pilot first approach is built to land.

AI native from day one. Demand responsive pricing, predictive staffing and footfall analytics are core product from launch, not a later add on. **Sold as a yield line, not a cost line.** Because verified identity makes it safe to open a private lot to outsiders on the owner's terms, ParkFlow can offer facility owners revenue from idle capacity, described in Section 6.1, rather than only a bill for better gates. That reframes the buying decision for a facilities manager and shortens hardware payback, a commercial difference, not just a technical one.

The installed base is a payment network. Each Edge Node is already an identity verifying, wallet debiting terminal. That gives ParkFlow a credible path into vehicle commerce beyond parking, described in Section 6.2, that a pure app layer aggregator, holding no site hardware, does not have.

7.3 Positioning at a glance

Capability	Manual token	ANPR only	FASTag only / app wallet	ParkFlow
Vehicle must stop	Yes	Usually not	Usually not	No
Fraud and tailgating resistance	Low	Medium	Medium	High
Onboarding friction for regulars	Low, but manual	Low	Medium, a new wallet or	Low, uses an existing

			app	wallet
Works for one off visitors	Yes	Partial	Partial	Yes, with a QR fallback
Ready for public sector or BOT deployment	Yes, but no data layer	Case by case	Case by case	Purpose built

Section 8. Visual Impact and User Journey

This section describes the before and after user journey in words. The original Stage 1 format calls for a visual representation here; a diagram based version is available on request.

Today: the manual grind

1. **Slow and stop.** The vehicle must brake and halt at the barrier.
2. **Take a token.** A manual paper or plastic token is issued by a guard.
3. **Queue up.** The driver waits behind other cars, especially at peak hours.
4. **Hunt for a slot.** The driver circles floors manually, looking for space.
5. **Pay and exit.** The driver finds the token, pays by cash or card, and waits for the barrier.

Typical total time: 8 to 12 minutes of active friction across a single round trip. Cash is handled. There is zero digital record.

With ParkFlow

1. **Drive up.** A normal speed approach, with no braking needed.
2. **Auto verified.** The RFID and ANPR cross check completes in under 300 milliseconds.
3. **Lane stays open.** The barrier lifts ahead of arrival, or there is no barrier at all.
4. **Guided to a slot.** LED signage and the app point straight to a free bay.
5. **Drive out, billed.** The fee is auto debited and an e receipt lands on the app.

Target total time: under 5 seconds of interaction per gate. Zero cash is handled. Every trip is fully logged.

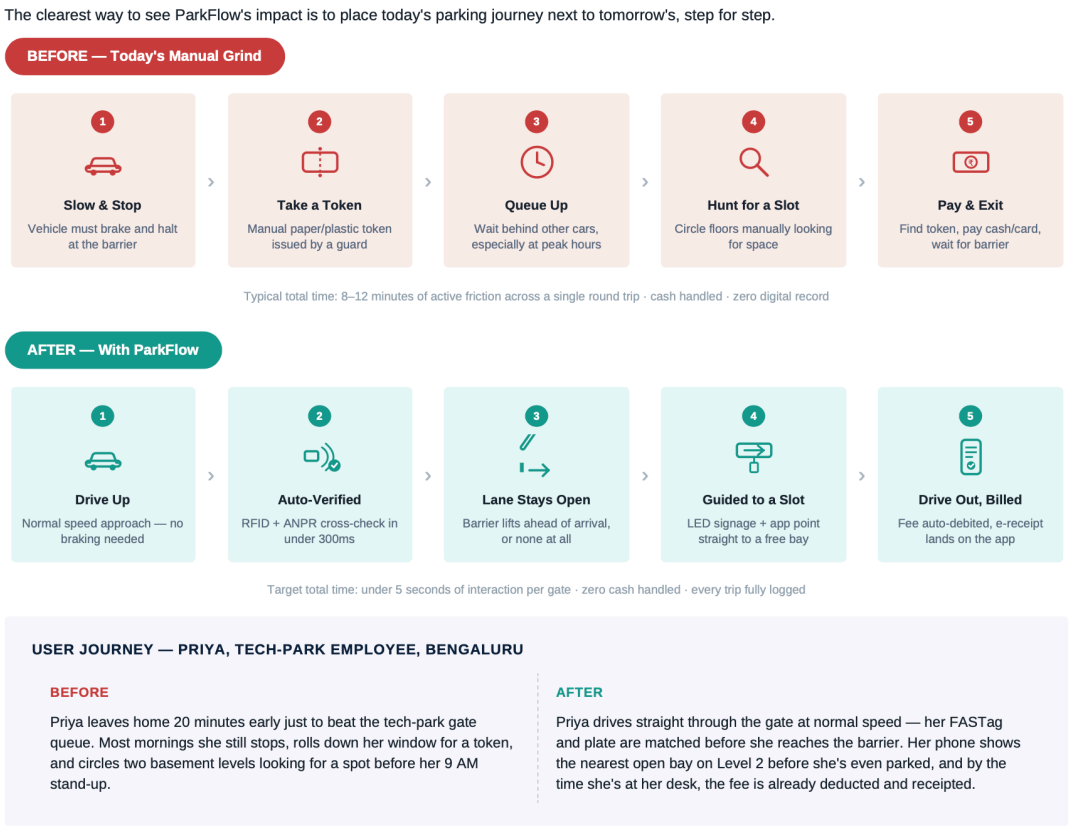
Illustrative impact: unlocking idle capacity

The table below illustrates, for a representative 1,200 bay tech park, how tenant demand leaves most of the structure idle outside working hours, and how a public after hours pool (Section 6.1) can convert that idle time into paid occupancy.

Time of day	Typical utilisation today	With a ParkFlow after hours pool
9 AM to 6 PM (working hours)	High, driven by tenant and employee demand	Unchanged. Tenant bays are untouched.
6 PM to 9 PM (early evening)	Low and falling	Moderate. Public bays open as tenants leave.
9 PM to 7 AM (overnight)	Very low	Meaningful, sold as overnight or event parking
Weekends and public holidays	Very low	Meaningful, sold as public day parking

This is an illustrative demand shape, not measured site data. It is included to make the commercial mechanism in Section 6.1 concrete. Actual idle windows and uplift will vary by facility, location and local demand.

The clearest way to see ParkFlow's impact is to place today's parking journey next to tomorrow's, step for step.



USER JOURNEY — PRIYA, TECH-PARK EMPLOYEE, BENGALURU

BEFORE

Priya leaves home 20 minutes early just to beat the tech-park gate queue. Most mornings she still stops, rolls down her window for a token, and circles two basement levels looking for a spot before her 9 AM stand-up.

AFTER

Priya drives straight through the gate at normal speed — her FASTag and plate are matched before she reaches the barrier. Her phone shows the nearest open bay on Level 2 before she's even parked, and by the time she's at her desk, the fee is already deducted and receipted.

Assumptions and Data Sources

FASTag adoption and NETC transaction volume figures are drawn from NHAI / NETC public statements and industry reporting current through 2025. Parking market sizing figures are drawn from IMARC Group's India Smart Parking Systems Market report and MarketsandMarkets' India Parking Management Market analysis, both 2025 editions.

Competitor information is based on public company disclosures, press coverage and funding announcements as of the most recent available reporting. Figures are self reported by the respective companies and are indicative, not audited.

Operational targets in Sections 4 and 5, including time per vehicle, throughput and revenue leakage reduction, are internal projections based on queueing theory estimates and comparable deployments. They are to be validated empirically during the Phase 0 pilot, not treated as guaranteed or audited results. The occupancy illustration in Section 8 is an illustrative weekday demand shape used to explain the commercial mechanism in Section 6.1. It is not measured site data.

The non parking use cases in Section 6.2 depend on NPCI merchant category approvals and acquirer partnerships that ParkFlow does not control. They are presented as optionality on already deployed hardware, not as committed roadmap items, and the core business case does not rely on them.